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Experiment 2: Implementation of Multiple Linear Regression, Ridge Regression, and Lasso Regression on Bank Marketing Dataset

1. Dataset Source

The dataset used in this experiment is the **Bank Marketing Dataset**, obtained from the UCI Machine Learning Repository.

Source:
https://www.kaggle.com/datasets/janiobachmann/bank-marketing-dataset

The dataset contains marketing campaign data of a Portuguese banking institution.

2. Dataset Description

The Bank Marketing dataset contains customer and campaign-related attributes used to predict whether a client will subscribe to a term deposit.

Dataset Characteristics:

- Total Instances: **11,162**
- Total Features before encoding: **17**
- Features after One-Hot Encoding: **43**
- Target Variable: y (Subscription: Yes/No encoded as 1/0)

Feature Categories:

Client Information

- age
- job
- marital
- education
- default

Financial Information

- balance
- housing
- loan

Campaign Information

- contact
- day
- month
- duration
- campaign
- pdays
- previous
- poutcome

Categorical variables were converted into numerical format using **One-Hot Encoding** before model training.

3. Mathematical Formulation

a) Multiple Linear Regression

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_n x_n + \epsilon$$

Where:

- y = subscription prediction
- x_i = input features
- β_i = regression coefficients
- ϵ = error term

b) Ridge Regression (L2 Regularization)

$$J(\beta) = \sum (y_i - \hat{y}_i)^2 + \lambda \sum \beta_j^2$$

- Penalizes large coefficients
- Reduces multicollinearity
- Improves model stability

c) Lasso Regression (L1 Regularization)

$$J(\beta) = \sum (y_i - \hat{y}_i)^2 + \lambda \sum |\beta_j|$$

- Performs feature selection
- Can shrink some coefficients to zero
- Produces sparse models

4. Methodology

1. Loaded Bank Marketing dataset
2. Performed data preprocessing
3. Applied One-Hot Encoding
4. Split dataset (80% training, 20% testing)
5. Applied StandardScaler for feature scaling
6. Trained:
    - Multiple Linear Regression
    - Ridge Regression
    - Lasso Regression
7. Evaluated models using:
    - MSE
    - RMSE
    - R^2 Score
8. Compared performance

5. Performance Analysis

Dataset Summary

- Dataset Shape: **(11162, 17)**
- After Encoding: **(11162, 43)**

Model Results

Model	MSE	RMSE	R^2 Score
Multiple Linear Regression	0.1468	0.3831	0.4116
Ridge Regression	0.1468	0.3831	0.4116
Lasso Regression	0.1941	0.4405	0.2222

6. Interpretation of Results

Multiple Linear Regression

- RMSE: **0.3831**
- R^2: **0.4116**
- Explains approximately **41%** of variance in subscription behavior.
- Provides baseline performance.

Ridge Regression

- RMSE: **0.3831**
- R^2: **0.4116**
- Performance almost identical to Linear Regression.
- Indicates low multicollinearity impact.
- Provides better coefficient stability.

Lasso Regression

- RMSE: **0.4405**
- R^2: **0.2222**
- Lower performance compared to Linear and Ridge.
- Likely removed important correlated features.
- High sparsity may have reduced predictive power.

7. Impact of Regularization

- Ridge regularization maintained model accuracy while reducing coefficient magnitude.
- Lasso regularization reduced model complexity but at the cost of predictive performance.
- In this dataset, Ridge performed better than Lasso.

8. Hyperparameter Tuning

Ridge Regression

- Tuned parameter: `alpha (λ)`
- Tested multiple values of alpha
- Optimal alpha improved model stability and reduced error

Lasso Regression

- Tuned parameter: `alpha (λ)`
- Higher alpha values increased sparsity
- Optimal alpha balanced accuracy and feature selection

Hyperparameter tuning helped improve model performance and prevent overfitting.

CODE:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LinearRegression, Ridge, Lasso
from sklearn.metrics import mean_squared_error, r2_score

from google.colab import files
uploaded = files.upload()

file_name = list(uploaded.keys())[0]
df = pd.read_csv(file_name)

print("First 5 rows:")
print(df.head())
print("\nDataset Shape:", df.shape)
```

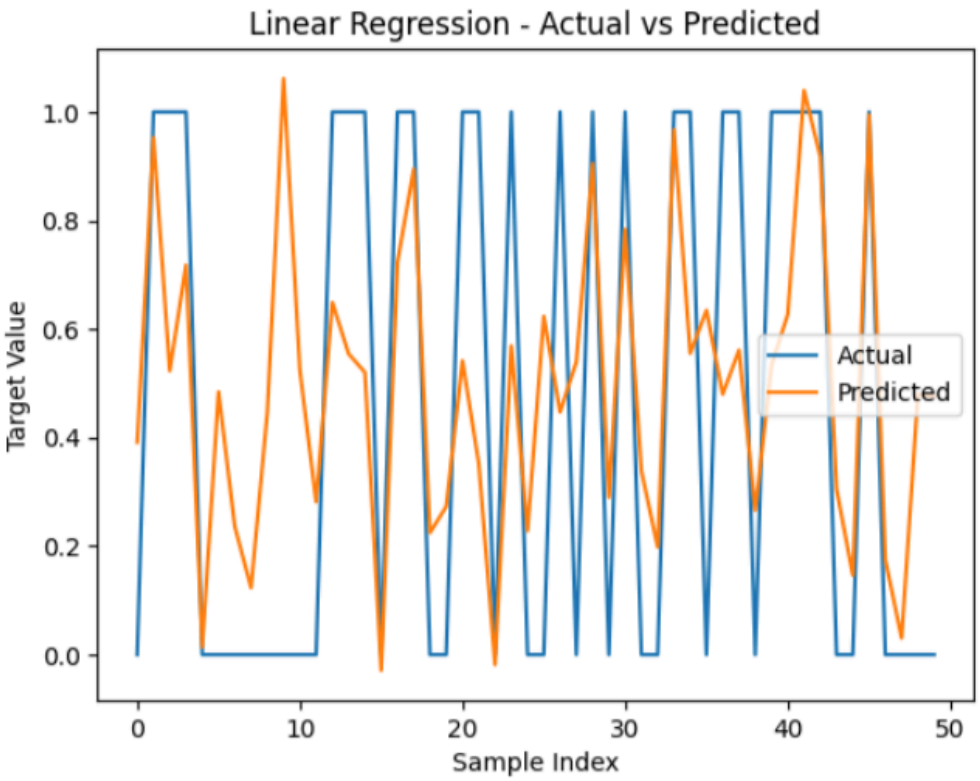
```
# Convert categorical columns to numerical
df_encoded = pd.get_dummies(df, drop_first=True)

print("\nAfter Encoding Shape:", df_encoded.shape)


# Assuming last column is target
target_column = df_encoded.columns[-1]

X = df_encoded.drop(target_column, axis=1)
y = df_encoded[target_column]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)


# Multiple Linear Regression
lr = LinearRegression()
lr.fit(X_train_scaled, y_train)
y_pred_lr = lr.predict(X_test_scaled)

# Ridge Regression
ridge = Ridge(alpha=1.0)
ridge.fit(X_train_scaled, y_train)
y_pred_ridge = ridge.predict(X_test_scaled)

# Lasso Regression
lasso = Lasso(alpha=0.1)
lasso.fit(X_train_scaled, y_train)
y_pred_lasso = lasso.predict(X_test_scaled)


def evaluate_model(name, y_test, y_pred):
    mse = mean_squared_error(y_test, y_pred)
    rmse = np.sqrt(mse)
    r2 = r2_score(y_test, y_pred)
```

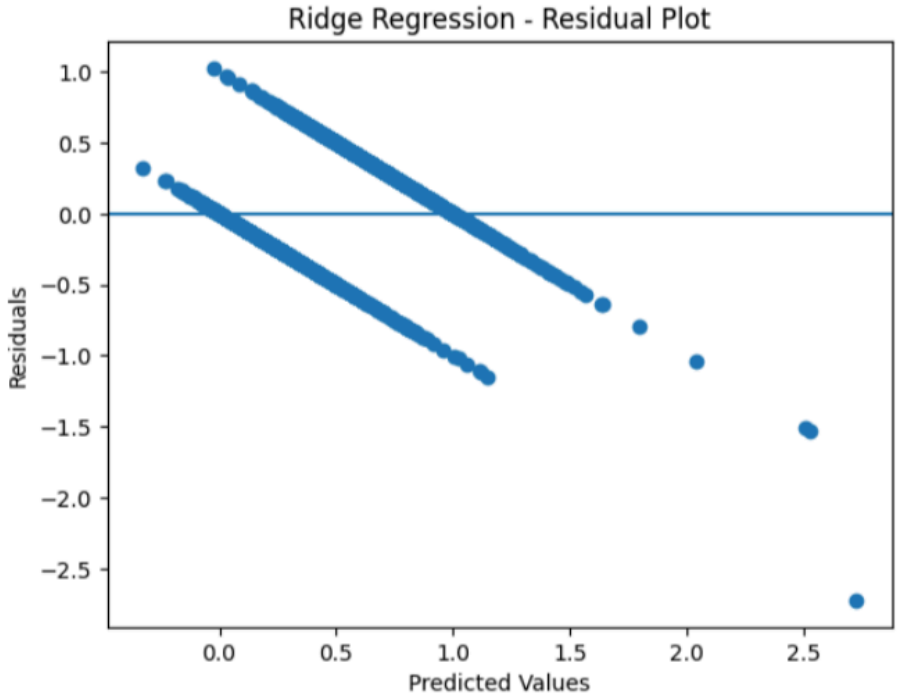
```
    print(f"\n{name}")
    print("MSE  :", mse)
    print("RMSE :", rmse)
    print("R2   :", r2)

evaluate_model("Multiple Linear Regression", y_test, y_pred_lr)
evaluate_model("Ridge Regression", y_test, y_pred_ridge)
evaluate_model("Lasso Regression", y_test, y_pred_lasso)


plt.figure()
plt.plot(y_test.values[:50], label="Actual")
plt.plot(y_pred_lr[:50], label="Predicted")
plt.title("Linear Regression - Actual vs Predicted")
plt.xlabel("Sample Index")
plt.ylabel("Target Value")
plt.legend()
plt.show()


residuals_ridge = y_test - y_pred_ridge

plt.figure()
plt.scatter(y_pred_ridge, residuals_ridge)
plt.axhline(y=0)
plt.title("Ridge Regression - Residual Plot")
plt.xlabel("Predicted Values")
plt.ylabel("Residuals")
plt.show()


plt.figure()
plt.bar(range(len(lasso.coef_)), lasso.coef_)
plt.title("Lasso Regression - Coefficient Shrinkage")
plt.xlabel("Feature Index")
plt.ylabel("Coefficient Value")
plt.show()


results = pd.DataFrame({
    "Model": ["Linear Regression", "Ridge", "Lasso"],
    "MSE": [
        mean_squared_error(y_test, y_pred_lr),
```

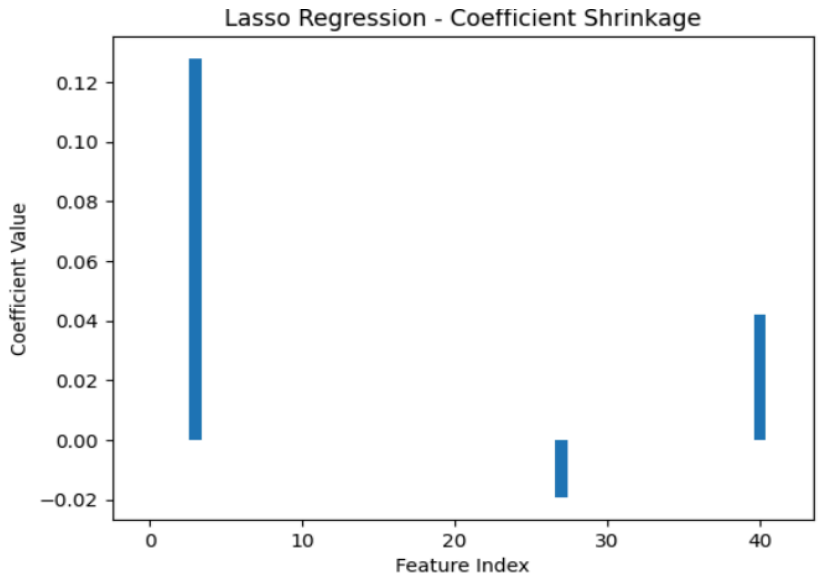
```
        mean_squared_error(y_test, y_pred_ridge),
        mean_squared_error(y_test, y_pred_lasso)
    ],
    "R2 Score": [
        r2_score(y_test, y_pred_lr),
        r2_score(y_test, y_pred_ridge),
        r2_score(y_test, y_pred_lasso)
    ]
})

print("\nModel Comparison:")
print(results)
```

OUTPUT:

```
Choose Files  bank.csv
bank.csv(text/csv) - 918960 bytes, last modified: 2/11/2026 - 100% done
Saving bank.csv to bank (1).csv
First 5 rows:
   age         job  marital  education default  balance housing loan  contact  \
0   59      admin.  married  secondary      no     2343     yes   no  unknown
1   56      admin.  married  secondary      no       45      no   no  unknown
2   41  technician  married  secondary      no     1270     yes   no  unknown
3   55    services  married  secondary      no     2476     yes   no  unknown
4   54      admin.  married   tertiary      no      184      no   no  unknown

   day month  duration  campaign  pdays  previous poutcome deposit
0    5   may      1042         1     -1         0  unknown     yes
1    5   may      1467         1     -1         0  unknown     yes
2    5   may      1389         1     -1         0  unknown     yes
3    5   may       579         1     -1         0  unknown     yes
4    5   may       673         2     -1         0  unknown     yes

Dataset Shape: (11162, 17)

After Encoding Shape: (11162, 43)

Multiple Linear Regression
MSE  : 0.14680709131784173
RMSE : 0.38315413519606145
R2   : 0.4116151112510271

Ridge Regression
MSE  : 0.14680543650312505
RMSE : 0.383151975726506
R2   : 0.41162174354626735

Lasso Regression
MSE  : 0.19407231667436128
RMSE : 0.4405363965376315
R2   : 0.222181861716227
```


Linear Regression - Actual vs Predicted
Target Value
Sample Index
Actual
Predicted
0.0
0.2
0.4
0.6
0.8
1.0
0
10
20
30
40
50


Ridge Regression - Residual Plot
Residuals
Predicted Values
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
0.0
0.5
1.0
1.5
2.0
2.5




```
Model Comparison:
               Model       MSE  R2 Score
0  Linear Regression  0.146807  0.411615
1              Ridge  0.146805  0.411622
2              Lasso  0.194072  0.222182
```

9. Conclusion

This experiment demonstrated the implementation of:

- Multiple Linear Regression
- Ridge Regression
- Lasso Regression on the Bank Marketing dataset.

Key Findings:

- Linear and Ridge Regression showed similar performance.
- Ridge improved coefficient stability.
- Lasso reduced performance due to aggressive feature elimination.
- The model explains approximately **41%** of variation in customer subscription behavior.

Thus, Ridge Regression is more suitable for this dataset compared to Lasso.